Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Projectiles

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A particle is projected with speed $20\,\text{m}\,\text{s}^{-1}$ at 30° to the horizontal. Find the initial horizontal and vertical components of velocity. **[3]**

2. A particle is projected at $25\,\text{m}\,\text{s}^{-1}$ at 40° to the horizontal. Find the time of flight and the range on horizontal ground ($g = 9.8$). **[5]**

3. A particle is projected at $30\,\text{m}\,\text{s}^{-1}$ at 45°. Find the maximum height and the range ($g = 9.8$). **[5]**

4. A particle is projected horizontally at $15\,\text{m}\,\text{s}^{-1}$ from a height of $20\,\text{m}$. Find the time to reach the ground and the horizontal distance travelled. **[5]**

5. A particle is projected at $10\,\text{m}\,\text{s}^{-1}$ at 60° to the horizontal. Find its speed and direction after $1\,\text{s}$ ($g = 9.8$). **[5]**

6. Find the equation of the trajectory of a particle projected with speed u at angle α to the horizontal. **[4]**

7. A particle is projected at $20\,\text{m}\,\text{s}^{-1}$ at 30°. Find the coordinates of the particle after $1.5\,\text{s}$ ($g = 9.8$). **[4]**

8. A particle is projected horizontally from a cliff $45\,\text{m}$ high at $12\,\text{m}\,\text{s}^{-1}$. Find the time of flight and the distance from the base of the cliff where it lands. **[5]**

9. A particle is projected at $40\,\text{m s}^{-1}$ at $30°$. Find the maximum height ($g = 9.8$). **[3]**

10. A particle is projected at $50\,\text{m s}^{-1}$ at $60°$. Find the range on horizontal ground ($g = 9.8$). **[4]**

11. A particle is projected at $15\,\text{m s}^{-1}$ at $45°$ from a point on horizontal ground. Find the time when it is at height $5\,\text{m}$. **[5]**

12. A particle is projected with speed u at angle α. Show that the time of flight is $\dfrac{2u\sin\alpha}{g}$. **[3]**

13. A particle is projected with speed u at angle α. Show that the range is $\dfrac{u^2\sin 2\alpha}{g}$. **[3]**

14. A particle is projected at $20\,\text{m s}^{-1}$ at $53°$ ($\sin 53° = 0.8$). Find the range ($g = 9.8$). **[4]**

15. A particle is projected horizontally at $8\,\text{m s}^{-1}$ from a height of $5\,\text{m}$. Find its speed on hitting the ground. **[5]**

Section B – Hard (Q16–25)

16. A particle is projected at $25\,\text{m s}^{-1}$ at $35°$ from the top of a $20\,\text{m}$ building. Find the time to reach the ground and the distance from the base of the building. **[7]**

17. A particle is projected at $30\,\text{m s}^{-1}$ at $40°$. Find the times when it is at a height of $10\,\text{m}$ and the horizontal distances at those times. **[6]**

18. A particle is projected from a point on a horizontal plane with speed $20\,\text{m s}^{-1}$ at angle α where $\tan\alpha = \frac{3}{4}$. Find the range and the maximum height. **[5]**

19. A particle is projected at $40\,\mathrm{m\,s^{-1}}$ at $30°$ towards a vertical wall 50 m away. Does it hit the wall before hitting the ground? If so, at what height? **[6]**

20. Show that the maximum range for a given speed u is $\dfrac{u^2}{g}$ and occurs when $\alpha = 45°$. **[4]**

21. A particle is projected at $15\,\mathrm{m\,s^{-1}}$ at $60°$ from a point 10 m above horizontal ground. Find the speed and direction on impact with the ground. **[6]**

22. A particle is projected with velocity $10\mathbf{i} + 15\mathbf{j}\,\mathrm{m\,s^{-1}}$ from the origin. Find the equation of the trajectory and the range. **[5]**

23. A particle is projected at speed u at angle α up a plane inclined at β to the horizontal. Write down the components of acceleration parallel and perpendicular to the plane. **[4]**

24. A particle is projected at $20\,\mathrm{m\,s^{-1}}$ at $45°$ from the foot of an inclined plane of angle $30°$. Find the range up the plane. **[7]**

25. A particle is projected from O with speed $30\,\mathrm{m\,s^{-1}}$ at $60°$. At the same instant another particle is projected from a point 40 m horizontally from O with speed $20\,\mathrm{m\,s^{-1}}$ at $45°$ towards the first. Do they collide? **[7]**

End of Worksheet • *Show all working clearly. Take* $g = 9.8\,\mathrm{m\,s^{-2}}$ *unless otherwise stated.*